

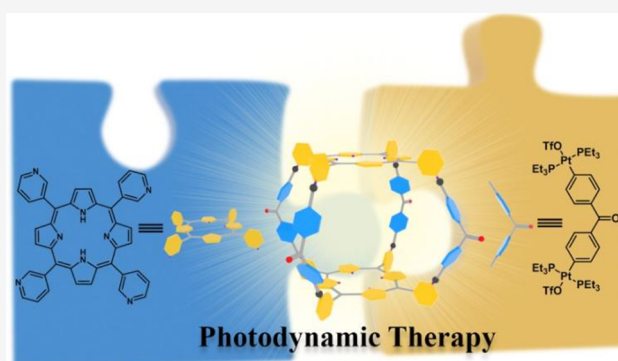
Self-Assembly of Porphyrin-Containing Metalla-Assemblies and Cancer Photodynamic Therapy

Xin Jiang,[▽] Zhixuan Zhou,[▽] Huang Yang, Chuan Shan, Hao Yu, Lukasz Wojtas, Mingming Zhang, Zhengwei Mao,^{*} Ming Wang,^{*} and Peter J. Stang^{*} Cite This: *Inorg. Chem.* 2020, 59, 7380–7388 Read Online

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ABSTRACT: In this report, we describe the synthesis of two porphyrin-containing Pt(II) supramolecular assemblies via coordination-driven self-assembly. X-ray crystallographic analysis on one assembly reveals that the metalla-assembly formation imposes large interchromophore distances, leading to a higher $^1\text{O}_2$ generation efficiency, relative to the corresponding small molecular precursors. The metalla-assemblies were examined as photosensitizers for photodynamic therapy as the potential reduction of the unfavorable self-aggregation phenomenon. In vivo and in vitro investigations demonstrate that the metalla-assemblies exhibit enhanced anticancer activity with minimal dose requirement and side effects comparable to the small molecule precursors. Thus, our work demonstrates that self-assembly provides a promising methodology for enhancing the therapeutic effectiveness of anticancer agents.



■ INTRODUCTION

Coordination-driven self-assembly is a powerful synthetic methodology for the preparation of discrete supramolecular coordination complexes (SCCs), including two-dimensional (2D) metallacycles and three-dimensional (3D) metalla-assemblies.¹ Via the selection of electron-rich organic donors and electron-poor metal acceptors, SCCs with predesigned, well-defined sizes and shapes can be obtained. As a result, the cavity size of the SCCs, as well as the distance and orientation of the functional groups incorporated within the SCCs, can be controlled, enabling precise functional modularity that leads to widespread applications in catalysis,² separation,³ and both light-emitting⁴ and polymeric materials.⁵

In the past few decades, photodynamic therapy (PDT) has been shown to be a promising clinical method for treating various types of cancer and infection.⁶ PDT requires a combination of three elements: photosensitizer (PS), light, and molecular oxygen. Activation of PS by irradiation of a suitable wavelength of light leads to the formation of reactive oxygen species (ROS), which induce cell death.⁷ Among the many PSs, porphyrin and its derivatives have attracted particular attention, because of their favorable photophysical properties.^{6b,8} However, porphyrin derivatives exhibit severe π – π stacking, because of their large planar structures, which facilitate aggregation and, thus, reduce their ROS generation efficiency.^{6b,7b,9} This characteristic has restrained the potential clinical applications of porphyrin derivatives and many other PSs. Although many novel carrier systems have been developed

to increase the solubility of porphyrin-based PSs, reduce aggregation,¹⁰ and provide efficient transport direct to cancer cells,¹¹ it is still challenging to achieve molecular level of separation of PSs in these systems, because of their strong molecular interactions.

Here, in order to regulate the aggregation of porphyrin-based PSs for improving anticancer performance, we design and synthesize two [2 + 4] metalla-assemblies, **1** and **2**, via the two-component coordination-driven self-assembly (Figure 1) between pyridyl-containing porphyrin motifs and a 120° di-Pt(II) motif. The two-component strategy provides structural simplicity and enhances stability compared to multicomponent systems that explored previously,^{8h} as shown by the nuclear magnetic resonance (NMR) studies and electrospray ionization mass spectrometry (ESI-MS), which is favorable for studies in complicated biological systems. 3D coordination geometry of metalla-assembly **1** was demonstrated by X-ray crystallographic analysis. The large, rigid structure of the metalla-assembly has accurate component proportion and high structural stability. It can efficiently increase the distance of two porphyrin units and inhibits the aggregation caused by molecular interaction, contributing to obtain a higher ROS

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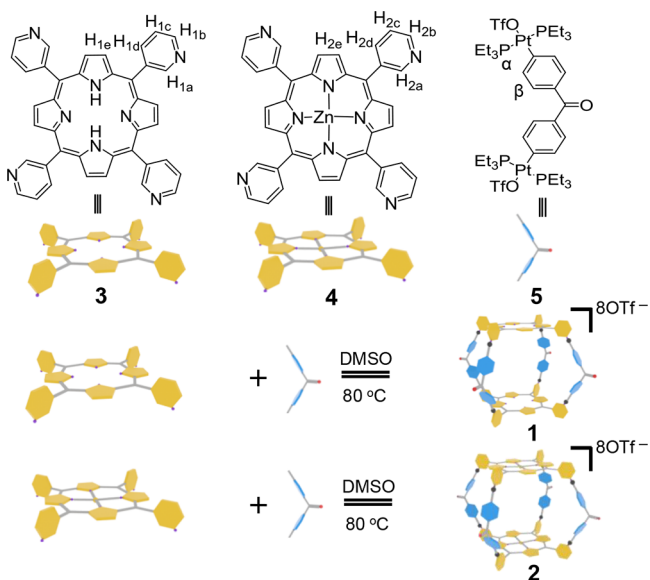


Figure 1. Self-assembly of supramolecular metalla-assemblies 1 and 2.

generation efficiency. In vitro and in vivo experiments were conducted to examine the anticancer activity of the metalla-assemblies, which revealed that both compounds exhibited potent anticancer activity with a low dose and minimal side effects. Thus, this work provides a promising methodology for efficient cancer treatment based on the design of supramolecular structures.

RESULTS AND DISCUSSION

Synthesis and Characterization of Metallo-Supramolecular Assemblies. The metalla-assemblies were synthesized via the self-assembly of 5,10,15,20-tetra (3-pyridyl) porphyrin (3) or zinc 5,10,15,20-tetra(4-pyridyl)-21*H*,23*H*-porphyrin (4) with a 120° dicationic platinum(II) motif (5) in a 1:2 stoichiometric ratio in dimethylsulfoxide (DMSO) at 80 °C for 12 h. The ^1H NMR of 1 and 2 both showed six sets of aromatic protons from the porphyrin units and the phenyl groups, which are consistent with the expected structures. All the protons corresponding to the pyridyl groups (H_{1a} to H_{1e}) exhibited downfield shifts upon the formation of 1, because of the loss of electron density upon coordination with platinum-metal centers, while the peak corresponding to the pyrrole H_{1f} shifted upfield (Figures 2a–e). This phenomenon is also

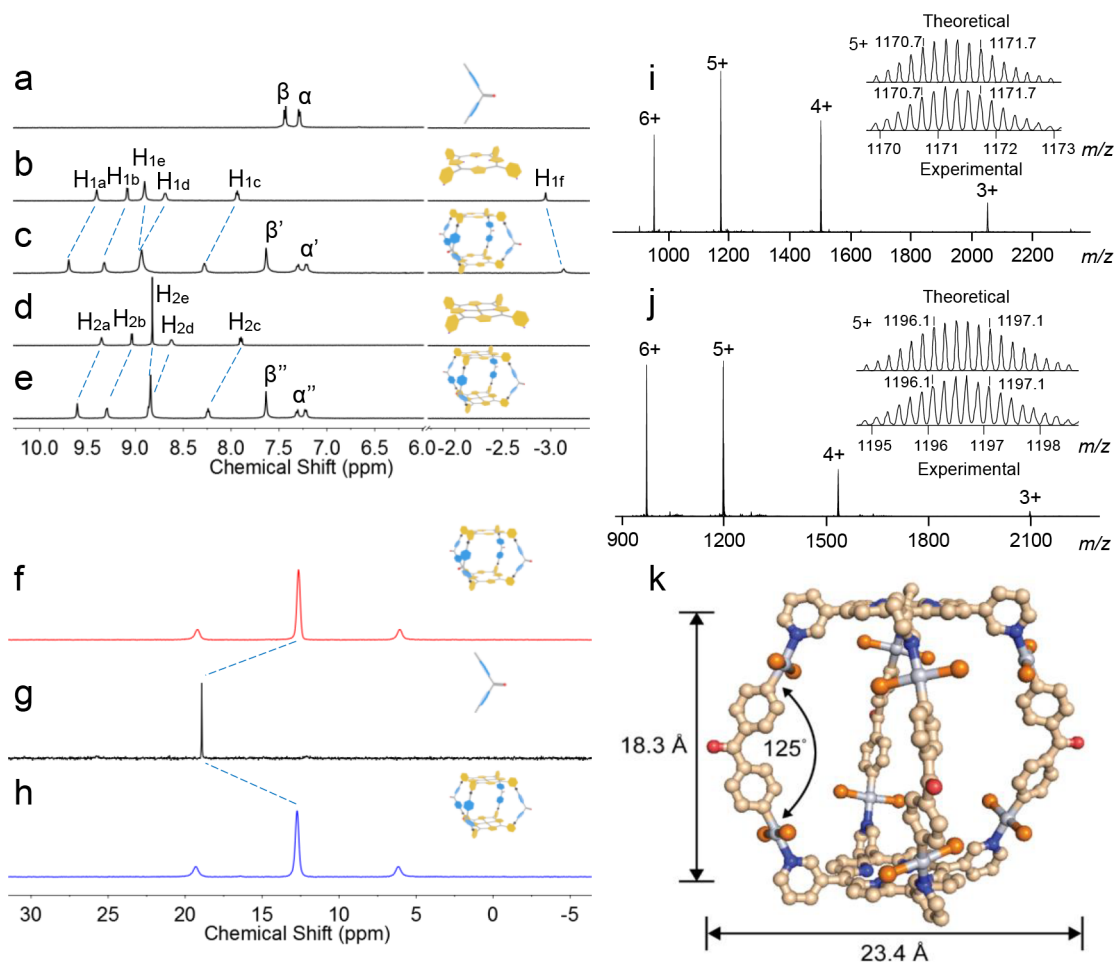


Figure 2. Structural characterization of metalla-assemblies. (a–e) Partial ^1H NMR and (f–h) partial $^{31}\text{P}\{^1\text{H}\}$ spectra (500 MHz, $\text{DMSO}-d_6$, 300 K) of 5 (spectra (a) and (g)), 3 (spectrum (b)), 1 (spectra (c) and (f)), 4 (spectrum (d)), and 2 (spectra (e) and (h)). (i, j) ESI-MS spectra of complexes 1 (panel (i)) and 2 (panel (j)). (k) X-ray crystal structure of 1. OTf anions and the ethyl groups on PEt_3 are omitted for the sake of clarity.

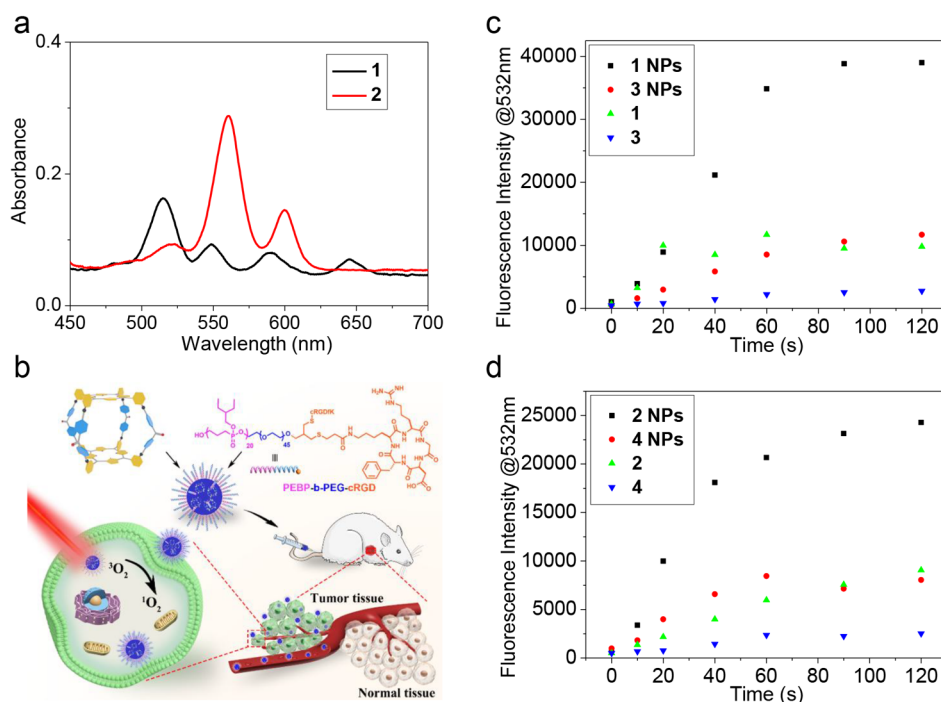


Figure 3. Photophysical research and cartoon representation of PDT. (a) UV-vis absorption spectra of **1** and **2** in DMSO ($c = 1 \times 10^{-5}$ M, 298 K). (b) Schematic diagrams of the **1**-cored nanoparticles (**1 NPs**) for cancer treatment. (c, d) Fluorescence intensity of the solution containing SOSG and different nanoparticles at 532 nm, as a function of irradiation time (638 nm, 0.5 W/cm²).

manifested in the ¹H NMR spectra of **2** and its ligand **4**. The ³¹P{¹H} NMR spectra of **1** and **2** both display a singlet with concomitant ¹⁹⁵Pt satellites ($\delta = 12.64$ ppm for **1** and 12.73 ppm for **2**, respectively) (see Figures 2f–h), which was shifted upfield, relative to the acceptor **5** ($\delta = 18.99$ ppm), because of the electron back-donation from the Pt atoms. The signals in the ³¹P{¹H} NMR, ¹H NMR, and 2D COSY NMR spectra (Figures S3–S12 in the Supporting Information) illustrate the formation of discrete and highly symmetric metalla-assemblies.

ESI-MS provides evidence for the formation of the metalla-assemblies. A series of peaks with different charge states from 3+ to 6+ were observed for both **1** and **2**, because of the loss of the triflate (OTf[−]) counterions. The isotope patterns of these peaks are in good agreement with the theoretical simulations (Figures 2i and 2j, as well as Figures S1 and S2 in the Supporting Information). After deconvolution of m/z , the molecular weights for **1** and **2** were determined to be 6600 and 6732 Da, respectively, confirming that these complexes are composed of two **3** or **4** and four diplatinum(II) acceptors (**5**).

Single crystals of **1** were obtained by slow vapor diffusion of diethyl ether into a nitromethane solution of the assembly (OTf[−] salt) over 4 weeks. In the solid state, 1·8OTf solvates crystallizes in monoclinic space group *P2₁/c* with a slight positional disorder, because of the flexible nature of the pyridyl groups in **3**. The assembly consists of two parallel porphyrins linked by four di-Pt(II) motifs. The di-Pt(II) motif adopts an angle of 125°, likely because of the slight twisting of the porphyrin group. The distance between the parallel Pt(II) building blocks is 23.4 Å, while the distance between the two porphyrin panels was found to be 18.3 Å (Figure 2k). The formations of ordered and complementary crystal arrangements are provided in the Supporting Information (Figure S13 and Table S1). As such, the metalla-assemblies formation via coordination efficiently separates the porphyrin groups,

prohibiting the tight packing that is undesirable for practical applications.

Photophysical Properties. The ultraviolet–visible light (UV-vis) absorption spectra of **1** and **2** were recorded in DMSO (see Figure 3a). Several absorption wavelengths were observed in the Q-band from 450 nm to 650 nm, which was attributed to the absorption transition between the ground state and the first excited singlet state (S₀–S₁). A broad absorption band centered at 638 nm was observed in **1**. The chelation of Zn ion in **2** led to hypsochromically shifted Q-bands, compared with **1**.¹² This indicates that **1** has more advantages over **2** in practical biological experiments, because of its stronger penetrating ability, as indicated by the redder absorption wavelength.

Fabrication of Supramolecular Assembly Loaded Nanoparticles. Since **1** and **2** are hydrophobic and not well-dispersed in the aqueous media, amphiphilic block polymer PEBP-*b*-PEG-*c*-RGD (see Figure 3b, as well as Figure S14 in the Supporting Information) was used to encapsulate them into the core of hydrophilic nanoparticles (denoted as **1**-cored nanoparticles (**1 NPs**) and **2**-cored nanoparticles (**2 NPs**), respectively), to enhance their colloidal stability and circulation time for further in vivo applications.¹³ Furthermore, the accumulation of nanoparticles in tumors can be expected with the collective effects based on the enhanced permeability and retention (EPR) effect and active targeting ability of cRGD moieties.¹⁴

The morphologies and sizes of the nanoparticles were investigated by transmission electron microscopy (TEM) and dynamic laser scattering (DLS). Spherical **1 NPs** with diameters ranging from 30 nm to 90 nm were observed in the TEM study (see Figure S15A in the Supporting Information). A slightly larger hydrophilic diameter was recorded from DLS (Figure S15B in the Supporting

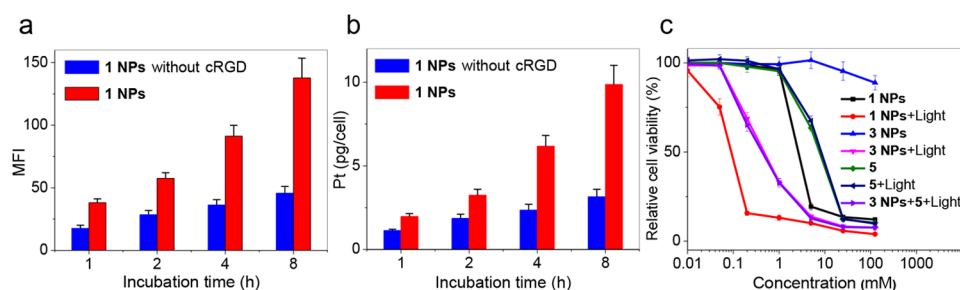


Figure 4. In vitro cell uptake and cytotoxicity of **1** NPs: (a) mean fluorescent intensity (MFI) and (b) intracellular platinum amount of 4T1 cells after treatment with the **1** NPs, with or without cRGD targeting moiety for different incubation times. (c) Relative cell viability of 4T1 cells treated with different administrations.

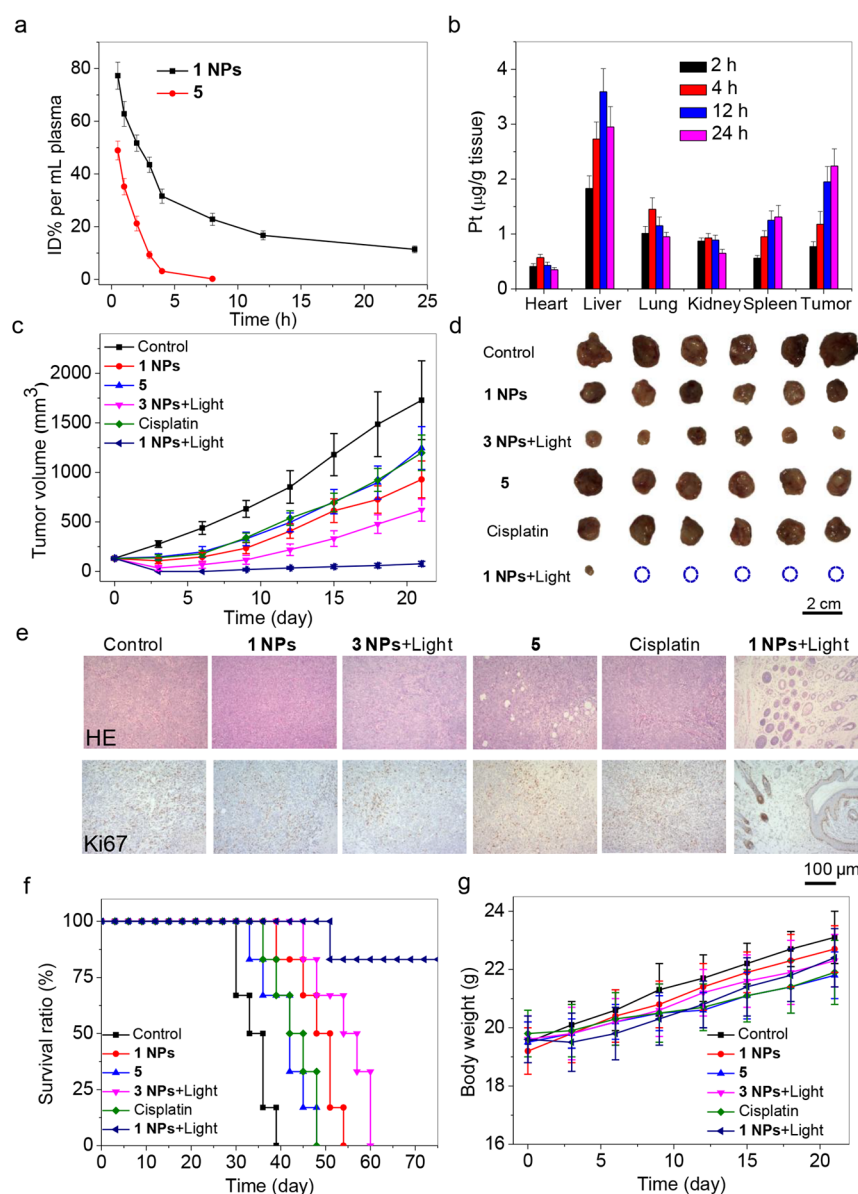


Figure 5. In vivo distribution and antitumor effect on 4T1 orthotopic breast cancer-bearing mice. (a) In vivo blood elimination kinetics of **5** and the **1** NPs at a dose of 2 mg Pt/kg body weight ($n = 4$). (b) Tissue distributions of the platinum in the main organs after iv injection of the **1** NPs at a dose of 2 mg Pt/kg body weight ($n = 4$) for different times. (c) Tumor volume as a function of time after different treatments ($n = 6$). (d) Photographs of 4T1 orthotopic tumors harvested from the mice receiving different treatments. (e) Hematoxylin and eosin (H&E) and Ki67 analyses of tumor tissues after various treatments. Ki67-positive tumor cells were stained brown. (Scale bar = 100 μm .) (f) Kaplan–Meier plots and (g) body weights of the mice bearing 4T1 tumors after different treatments. The irradiation density was 0.5 W/cm² at 638 nm for 6 min. Data are expressed as mean $\pm$ standard error of measurement (SEM). [The double asterisk symbol (**) denotes $P < 0.01$.]

Information), because of the hydration of the nanoparticles (NPs). An enhancement in diameter from 35.7 nm to 61.8 nm was observed after loading **1** into the nanoparticles, indicating the successful encapsulation of the assemblies by the amphiphilic polymers. Furthermore, the zeta potential of the nanoparticles increased from -46.9 mV to -5.4 mV upon encapsulation of **1** (see Figure S15C in the Supporting Information), suggesting that the electrostatic interaction is one of the major driving forces in the formation of self-assemblies, in addition to hydrophobic interaction. The **1** NPs are stable in phosphate-buffered saline (PBS) containing 10% fetal bovine serum at 37°C over 48 h of incubation (see Table S2 in the Supporting Information), demonstrating the promising colloidal stability of **1** NPs in biological milieus.

Singlet oxygen sensor green (SOSG) was used to quantify the ROS generation of the nanomaterials upon laser irradiation (638 nm, 0.5 W/cm²). We used the same molar concentration of porphyrin units to compare the ROS production ability between metalla-assembly (250 nM) and porphyrin (500 nM) alone. As shown in Figure 3c, low fluorescence enhancement was observed from **3** and **3**-cored nanoparticles (**3** NPs), because of the poor ROS generation efficiency induced by strong π - π stacking interactions. In contrast, a 4-fold enhancement for the SOSG fluorescence is observed upon the introduction of cage structure (**1** NPs), suggesting a high photosensitizing efficacy. The high yield of ROS generation from **1** NPs can be attributed to the supramolecular metalla-assembly structure to efficiently prevent the aggregation of porphyrin units and the incorporation of platinum atoms in the assembly. This is beneficial for the generation of reactive singlet oxygen species by promoting intersystem crossing.¹⁵ Compared to **1** NPs, incorporation of Zn ions in **2** NPs leads to lower ROS generation efficiency under laser irradiation, because of the hypsochromically shifted Q-bands (see Figure 3d).

In Vitro Cytotoxicity. The presence of cRGD within **1** NPs should endow the nanoparticle with tumor targeting capability.¹⁶ Receptor-mediated uptake of the **1** NPs in mouse triple-negative breast cancer cells (4T1) was quantitatively confirmed by flow cytometry and inductively coupled plasma mass spectrometry (ICP-MS) (see Figures 4a and 4b). Faster and higher cellular internalization of **1** NPs is observed relative to the nanoparticles without cRGD modification.

The cytotoxicity of the compounds in darkness and when subjected to light irradiation (+Light) was quantified using 3-(4',5'-dimethylthiazol-2'-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay (see Figure 4c, as well as Table S3 in the Supporting Information). The Pt(II) acceptor displayed moderate cytotoxicity, with half-maximal inhibitory concentration in darkness (IC₅₀) values of 7.85 ± 0.8 μM and 8.15 ± 0.7 μM for **5** and **5**+Light, respectively. These values are higher than that of commercially available cisplatin (IC₅₀ = 4.39 ± 0.7 μM). This result also suggests that light irradiation on **5** cannot enhance its toxicity. In contrast, **3** NPs did not show obvious cytotoxicity without laser irradiation, while the half-maximal inhibitory concentration with light irradiation (LD₅₀) of **3** NPs was determined to be 0.51 ± 0.12 μM upon light irradiation (638 nm for 2 min, 0.2 W/cm²). The most encouraging results were observed from the **1** NPs+Light group, showing an LD₅₀ value of 0.087 ± 0.009 μM upon irradiation, which was much lower than those of the small molecular precursors. From our previous study, we found the SCCs can interact with DNA, because of the binding effect between Pt and DNA bases.^{4h} It

is possible that the assemblies degrade in a biological environment and release Pt fragments and show anticancer actions. The phototoxicity index (PTI), which defines the ratio of the IC₅₀ in darkness and LD₅₀ under light irradiation, was calculated to be 40.7. It is quite difficult to compare the PDT performance based on PTI directly, because the lasers are different. At least the PTI of **1** NPs is comparable to aminolaevulinic acid (ALA), which is the porphyrin prodrug used in clinics.¹⁷ The results suggested the potential of the **1** NPs as an efficient PS for PDT treatment. Annexin V-FITC/PI assay (Figure S16 in the Supporting Information) and fluorescence in diacetate/propidium iodide (FDA/PI) costaining assay (Figure S17 in the Supporting Information) also suggest that **1** NPs possesses potent PDT effect.

In Vivo Distribution. In order to study their pharmacokinetics, the **1** NPs and free **5** were intravenously (iv) injected into the mice at a dose of 2 mg/kg Pt. Blood samples were collected at various time points after injection. By quantifying the blood platinum concentration using ICP-MS, the blood circulation half-life of the **1** NPs was calculated to be 2.39 ± 0.4 h, which was 4.5 times that of **5** (Figure 5a). Approximately 11.4% **1** NPs of the injected dose (ID) remained in the plasma at 24 h, while **5** was almost completely eliminated from the bloodstream 8 h post-injection.

The area under the curve (AUC) of the **1** NPs significantly increased to 149 $\mu\text{g mL}^{-1}$ h, which is 18.9-fold higher than that of **5** (7.86 $\mu\text{g mL}^{-1}$ h). These results suggest that **1** NPs possess a size favorable for long circulation and selective accumulation in the tumor site, which may lead to improved therapeutic efficacy and reduced side effects. Prolonged circulation of nanoparticles in the bloodstream is indispensable for successful targeted delivery and efficient treatment, which was achieved by surface-grafting of PEG shells, thereby preventing them from being cleared by the reticuloendothelial system (RES).¹⁸

To acquire a quantitative assessment of the biodistribution profile of the **1** NPs, the platinum contents of different organs were analyzed by ICP-MS (Figure 5b). Biodistribution evaluations indicated that the **1** NPs effectively accumulated in the tumor, approaching 2.24 ± 0.31 $\mu\text{g/g}$ tissue 24 h post-administration, which was significantly higher than that of **5** (0.39 ± 0.05 $\mu\text{g/g}$ tissue; see Figure S18 in the Supporting Information). A high concentration of **1** NPs can be observed in the liver, which was attributed to the RES capture and subsequent excretion. Precursor **5** showed a significantly different distribution in major organs, compared to that of **1** NPs, mainly located at the kidney. These findings suggest that **1** NPs were more widely available for tumor uptake than small-molecule drugs, while also reducing the adverse effects toward normal tissues.

In Vivo PDT. In vivo antitumor efficacy of the **1** NPs was evaluated on the highly aggressive 4T1 orthotopic breast cancer-bearing mice. 4T1 orthotopic tumor-bearing mice with tumor volume around 130 mm³ were randomly divided into six groups, and administered with physiological saline, cisplatin, **5**, **3** NPs+Light, **1** NPs, and **1** NPs+Light, respectively. A laser with low power density (0.5 W/cm²) and optimized irradiation time (6 min) was chosen for PDT at 24 h post-iv-injection. No noticeable skin burning was observed by application of the laser alone. The tumors of the mice administered with saline grew rapidly (see Figure 5c), exhibiting a 13.0-fold increase of tumor volumes after 21 days, compared to their original volume. The growth of tumors was slightly reduced by

cisplatin, **5**, or **1** NPs. Although the tumor size was notably reduced by the PDT in the first 6 days for the group subjected to **3** NPs+Light, a rapid recovery of tumor growth was evident afterward, likely because of the relatively low ROS generation of **3** NPs that was unable to clear all cancer cells distal from the primary tumor. **1** NPs+Light displayed the highest antitumor efficacy among these groups and almost completely eradicated tumors without recurrence (5 out of 6 mice) in the experimental period. The tumors were excised after 21 days post-treatment, and the weights of the tumors were assessed (see Figure Sd, as well as Figure S19 in the Supporting Information). The tumor growth inhibition ratio of the groups treated with **1** NPs upon light irradiation was 98.4%, while those for **1** NPs, **5**, **3** NPs+Light, and cisplatin were 51.5%, 37.3%, 60.5%, and 31.2%, respectively. These results demonstrated the antitumor efficacy of **1** by laser-irradiation-activated PDT to ablate the tumor without recurrence after a single treatment.

Immunohistochemical analyses were supportive of the results concerning tumor inhibition discussed above (Figure 5e). Hematoxylin and eosin (H&E) stain revealed that tumors in the control groups showed varying levels of recession, compared with the saline-treated group. For the mice treated with **1** NPs+Light, no tumor cells were observed, suggesting successful destruction of the tumor. Ki67-positive immunohistochemical staining further revealed the lowest proliferation (brown spots) in the tumor area from the photochemotherapy group (**1** NPs+Light).

The systemic toxicity of the nanomedicine was evaluated using body weight loss and survival rate as indications. For **5** and cisplatin administrations, the body weight of mice decreased within the first week, because of their systematic toxicity and related side effects (Figure 5g). Histological analyses provided insight into the systemic toxicity caused by **5** or cisplatin. Some pulmonary and hepatic damage was observed from the mice in **5** in cisplatin groups (see Figures S20 and S21 in the Supporting Information). In contrast, no apparent changes in body weight and histological examination were observed from the mice receiving other treatments (see Figures S22–S24 in the Supporting Information), implying that **1** NPs+Light has minimal systemic toxicity to mice after a single injection. The median survival rates for the mice treated with saline, **1** NPs, **5**, **3** NPs+Light, and cisplatin were calculated to be 36, 49, 42, 54, and 42 d, respectively; whereas PDT using **1** NPs+Light greatly prolonged mice survival over 75 days with a single death (Figure 5f). These results demonstrated PDT using **1** NPs+Light effectively prolonged mice lifespans without obvious side effects. A positive therapeutic effect was also observed on the cisplatin-resistant tumor model in vitro and in vivo (see Figures S25 and S26 in the Supporting Information).

The lung tissues were collected and investigated after the mice were sacrificed (see Figure S27A in the Supporting Information) to study the antimetastatic effect of **1** NPs. The average metastatic tumor nodules per lung were counted to be 5.67, 2.17, 3.67, 1.17, and 4.2 for the mice treated with saline, **1** NPs, **5**, **3** NPs+Light, and cisplatin, respectively (see Figure S27B in the Supporting Information). The tumor coverage percentages of the lung surface were calculated to be 9.33%, 2.23%, 4.37%, 0.83%, and 5.51% for the mice treated with saline, **1** NPs, **5**, **3** NPs+Light, and cisplatin, respectively (see Figure S27C in the Supporting Information). In contrast to these groups, only one tumor nodule was visualized from the

six lungs of mice that received the combinational treatment using **1** NPs+Light. The average tumor coverage on the lung surface was only 0.07%, indicating strong antimetastatic efficacy of the PDT by **1** NPs.

CONCLUSIONS

In summary, we designed and synthesized two supramolecular metalla-assemblies using the coordination-driven self-assembly between porphyrin derivatives (**3** and **4**) and 120° bimetallic Pt(II) acceptors (**5**). The single crystal of **1** indicates that the formation of metalla-assemblies efficiently inhibits the intramolecular and intermolecular stacking interactions of the porphyrin motifs, leading to a higher ROS generation efficiency, relative to the precursor. **1** NPs and **2** NPs were prepared by encapsulating the metalla-assemblies into the core of hydrophilic nanoparticles by using amphiphilic block polymer PEBP-*b*-PEG-*c*RGD, which enhances their colloidal stability and prolongs their circulation time. **1** NPs showed an IC₅₀ value of 0.087 ± 0.009 μM upon light irradiation, which was significantly lower than those of the small molecule precursors. In vivo studies suggest that PDT **1** NPs exert tumor inhibition effects without recurrence and metastasis after a single treatment against orthotopic breast cancers and drug-resistant cancers. This research provides a promising strategy for efficient cancer treatment by modulating the properties of functional motifs using self-assembly.

EXPERIMENTAL SECTION

General Procedures. Compounds **5**,¹⁹ **4**,²⁰ and PEBP-*b*-PEG-*c*RGD^{13c,11c} were prepared according to the reported literature. All reagents were purchased from Sigma–Aldrich, Matrix Scientific, and Alfa Aesar, respectively, and were used without further purification. Column chromatography was conducted using basic Al₂O₃ (Brockman I, activity, 58 Å) or SiO₂ (VWR, 40–60 μm, 60 Å), and the separated products were visualized using UV light. NMR spectra data were recorded on a 500-MHz Bruker Model Avance NMR spectrometer in CDCl₃, DMSO-*d*₆, with tetramethyl silane (TMS) as reference. ESI-MS was recorded with a Waters Synapt G2 tandem mass spectrometer, using solutions of 0.5 mg of sample in 1 mL of DMSO/MeOH/acetone (1:1:3, v/v) for complexes. The X-ray diffraction (XRD) data were measured on a Bruker Model D8 Venture PHOTON II CPAD system that was equipped with a Cu Kα INCOATEC ImuS microfocus source (λ = 1.54178 Å). The TEM images of the drop-cast samples were taken with a JEOL Model 2010 TEM system.

Synthesis of Complex 1. **3** (10.0 mg, 16.2 μmol) and **5** (43.4 mg, 32.4 μmol) were weighed accurately and dissolved in 2.5 mL of DMSO, and the mixture was heated at 80 °C for 10 h. After cooling to room temperature, an excessive amount of diethyl ether was added and a precipitate was formed, which was filtered and washed with diethyl ether, then dried in vacuo to obtain a dark brown solid (51.5 mg, 96.4%). ¹H NMR (500 MHz, DMSO-*d*₆, 300 K, ppm): δ 9.60 (m, 8H, Py-*H*^a), δ 9.33 (d, 8H, Py-*H*^b), δ 8.94 (m, 24H, Py-*H*^d, P-*H*^e), δ 8.28 (t, 8H, Py-*H*^c), δ 7.63 (d, 16H, Ph-*H*²), δ 7.20–7.30 (m, 16H, Ph-*H*¹), δ –3.13 (s, 4H, P-*H*^f). ¹³C DEPTQ NMR (500 MHz, DMSO-*d*₆, 300 K, ppm): δ 194.01, 154.46, 153.24, 143.76, 142.94, 140.53, 137.29, 135.39, 131.93, 130.08, 128.92, 127.06, 122.01, 119.86, 115.42, 42.57, 41.80, 40.89, 12.65, 7.89. ESI-MS (*m/z*): 2049.2 [M-3OTf]³⁺ (calcd *m/z*: 2049.2), 1500.9 [M-4OTf]⁴⁺ (calcd *m/z*: 1500.9), 1170.7 [M-5OTf]⁵⁺ (calcd *m/z*: 1170.7), 950.6 [M-6OTf]⁶⁺ (calcd *m/z*: 950.6).

Synthesis of Complex 2. **4** (10 mg, 14.7 μmol) and **5** (39.4 mg, 29.4 μmol) were weighed accurately and dissolved in 2.5 mL of DMSO, and the mixture was heated at 80 °C for 10 h. After cooling to room temperature, an excessive amount of diethyl ether was added and a precipitate was formed, which was filtered and washed with

diethyl ether, then dried in vacuo to get a dark purple solid (47.0 mg, 95.1%). ^1H NMR (500 MHz, $\text{DMSO}-d_6$, 300 K, ppm): δ 9.60 (s, 8H, $\text{Py}-\text{H}^a$), δ 9.29–9.30 (d, 8H, $\text{Py}-\text{H}^b$), δ 8.84–8.87 (m, 24H, $\text{Py}-\text{H}^d$, $\text{ZP}-\text{H}^e$), δ 8.22–8.25 (t, 8H, $\text{Py}-\text{H}^c$), δ 7.64 (d, 16H, $\text{Ph}-\text{H}^2$), δ 7.21–7.32 (m, 16H, $\text{Ph}-\text{H}^1$). ^{13}C DEPTQ NMR (500 MHz, $\text{DMSO}-d_6$, 300 K, ppm): δ 193.98, 154.08, 152.81, 150.01, 143.60, 142.80, 141.78, 137.27, 135.40, 132.49, 131.92, 130.10, 128.90, 126.72, 122.27, 119.71, 115.53, 41.81, 12.64, 7.88. ESI-MS (m/z): 2093.2 [$\text{M}-3\text{OTf}$] $^{3+}$ (calcd m/z : 2093.2), 1532.4 [$\text{M}-4\text{OTf}$] $^{4+}$ (calcd m/z : 1532.4), 1196.1 [$\text{M}-5\text{OTf}$] $^{5+}$ (calcd m/z : 1196.1), 971.8 [$\text{M}-6\text{OTf}$] $^{6+}$ (calcd m/z : 971.8).

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.9b02775>.

Synthetic and characterization data (NMR, ESI-MS, in vitro and in vivo experiment) (PDF)

Accession Codes

CCDC 1950332 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Author Contributions

[†]These authors contributed equally.

Notes

The authors declare no competing financial interest.

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